ECS [Elastic container service]

→ Containers are deployed on EC2 instances
→ AWS handles deployment
→ Use ECS agent to communicate

Launch type

Manual	Fargate
- You maintain EC2 instance infrastructure & Auto-scaling Group	- Serverless, automatic scale up/down.
- You calculate CPU utilization, memory utilization etc	- On demand, you just specify how much CPU power you need & it autoscales.
- You pay even if EC2 instance are Idle	- On demand only

- Manual but very Easy with help of AWS

ECS - ~~IAM~~ IAM roles

ECS Instance Role

- IAM policy for the ECS agent
- Give all permission needed to the agent

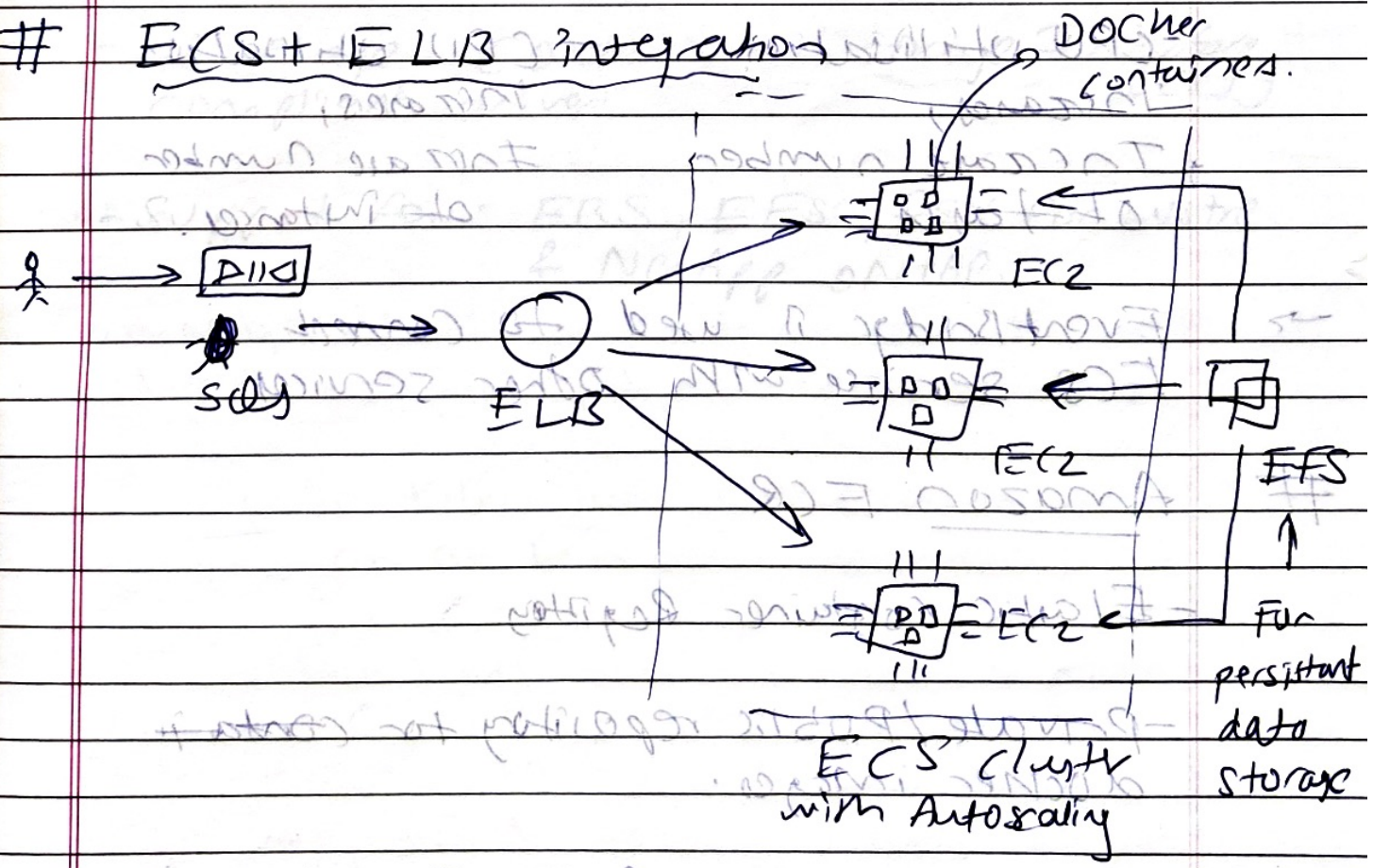
ECS Task Role

- IAM roles on basis of service/Task

Eg- Task 1
- Store to S3, DynamoDB etc

Task-2

ECS + ELB Integration



→ Fargate + EFS = Serverless + persistent data storage.

→ ECS have a cluster capacity provider that manage & autoscale when needed.

→ Task definition in ECS: (from Dockerfile)

It tells ECS - what container to run?
- How to run it?
- What resources it need?
↳ networking, ELB, Autoscaling etc

[manual or Forgot]

→ Autoscaling

ECS service Autoscaling	EC2 instance Autoscaling
- CPU utilisation increases,	- CPU utilisation increases,
- Increase number of tasks	Increase number of instances.

→ EventBridge is used to connect ECS service with other services.

Amazon ECR

- Elastic Container Registry

- Private / Public repository for ~~conta~~ docker images.

ECR → Pull image
↓ Deployed on
ECS.

Amazon EKS [Elastic Kubernetes Service]

- Managed by Kubernetes + AWS
- Gives standardization & more control to deployment
- Can be deployed on EC2 instance OR Fargate
- Can use Autoscaling features.
- Gives more control over deployment using Kubernetes.

→ EKS Data Volumes

- Leverage Container Storage Interface [CSI] compliant driver
- Support for EBS, EFS, FSX for Lustre & NetApp ONTAP.